

COMBINATORICS

EXAM 1

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1. (25pts): Let a_n be the number of ways to partition an n -element set into ordered pairs (i.e. matchings where order within each pair matters). Find the exponential generating function for a_n . Derive a closed-form expression for a_n .

Solution.

□

2. (25pts): Let a_n be the number of ways to distribute n labeled objects into any number of labeled boxes such that each box contains at least two objects. Construct the exponential generating function for a_n . Extract a formula for a_n .

Solution.

□

3. (25pts): The Catalan numbers are defined by $C_n = \frac{1}{1+n} \binom{2n}{n}$. Prove that C_n satisfies the recurrence $C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$. Give a combinatorial interpretation of this recurrence.

Proof.

□

4. (25 pts): How many integers between 1 and 1000 (inclusive) are divisible by at least one of 3, 5, or 7? Use the Inclusion-Exclusion Principle and show all steps.

Solution.

□

5.

6.

7.

8.